

Programming for AI

BS(Artificial Intelligence) Part-2- 2nd Semester

Prof. Dr. Imtiaz Korejo

Python Loops

Week- , Lecture-





Outline of Today's Lesson

1. Python While Loops

1. The While loop
2. The Break Statement
3. The Continue Statement
4. The else Statement in Loops

2. Python For Loops

1. For Loop
2. Looping through a String
3. Break Statement
4. Continue Statement
5. The range() Function
6. Else in For Loop

7. Nested Loops

8. The Pas Statement

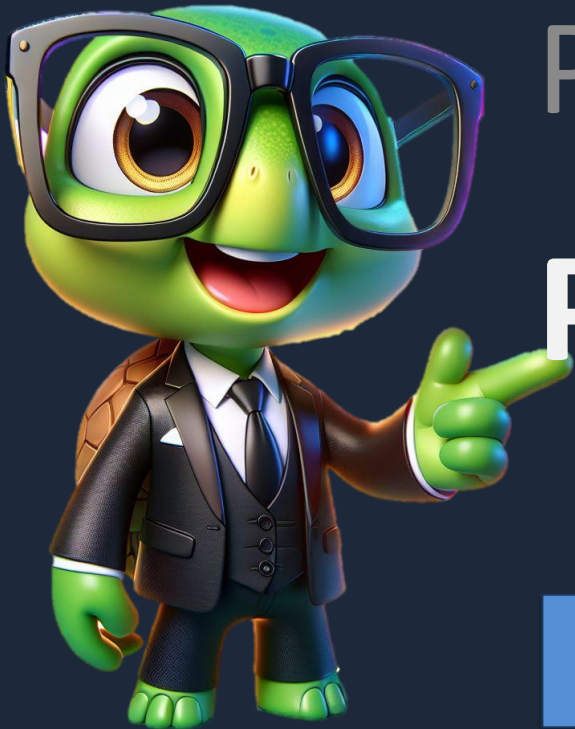
9.



Section-1

Python Programming

PYTHON WHILE LOOP



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The While Loop

- With the **while** loop we can execute a set of statements as long as a condition is true
- Print i as long as i is less than 6:

```
File Edit Selection View Go Run ...  
Example_2.py Example_3 Example_4.py Example_5.py  
  
i = 1  
while i < 6:  
    print(i)  
    i += 1  
  
Microsoft Windows [Version 10.0.22631.5624]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\Dr Imtiaz Korejo\Desktop\pythonworkspace>  
  
1  
2  
3  
4  
5
```





The break Statement

- With the **break** statement we can stop the loop even if the while condition is true:
- Exit the loop when i is 3:

```
File Edit Selection View Go Run ...  
Example_2.py Example_3 Example_4.py Example_5.py  
  
i = 1  
while i < 6:  
    print(i)  
    if i == 3:  
        break  
    i += 1  
  
Microsoft Windows [Version 10.0.22631.5624]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\Dr Imtiaz Korejo\Desktop\pythonworkspace>  
1  
2  
3
```



The Continue Statement

- With the **continue** statement we can stop the current iteration, and continue with the next:
- Continue to the next iteration if i is 3:

```
File Edit Selection View Go Run ...  
Example_2.py Example_3 Example_4.py Example_5.py  
  
i = 0  
while i < 6:  
    i += 1  
    if i == 3:  
        continue  
    print(i)  
  
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS  
Microsoft Windows [Version 10.0.22631.5624]  
(c) Microsoft Corporation. All rights reserved.  
C:\Users\Dr Imtiaz Korejo\Desktop\pythonworkspace>  
  
1  
2  
4  
5  
6
```





The else Statement

- With the **else** statement we can run a block of code once when the condition no longer is true:
- Print a message once the condition is false:

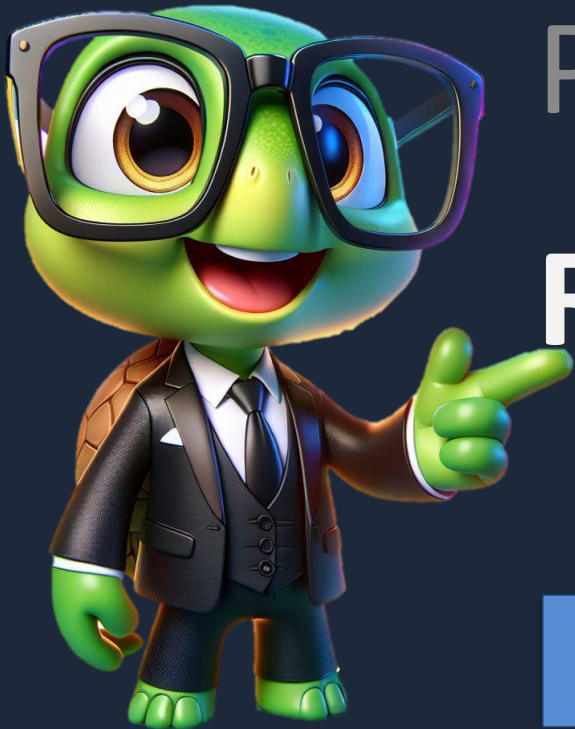
```
File Edit Selection View Go Run ...  
Example_2.py Example_3 Example_4.py Example_5.py  
  
i = 1  
while i < 6:  
    print(i)  
    i += 1  
else:  
    print("i is no longer less than 6")  
  
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS  
  
Microsoft Windows [Version 10.0.22631.5624]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\Dr Imtiaz Korejo\Desktop\pythonworkspace>  
  
1  
2  
3  
4  
5  
i is no longer less than 6
```



Section-2

Python Programming

REAL WORLD EXAMPLES



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1. Data Cleaning Loop: Removing Invalid Data

#Problem 1:

#ATM PIN verification (3 tries allowed)

correct_pin = 1234

attempts = 0

while attempts < 3:

pin = int(input("Enter your PIN: "))

if pin == correct_pin:

print(" Access granted.")

break

else:

print("Wrong PIN.")

attempts += 1

if attempts == 3:

print("Card blocked. Too many wrong attempts.")

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Microsoft Windows [Version 10.0.22631.5624]
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C:\Users\Dr Imtiaz Korejo\Desktop\pythonworkspace>

Enter your PIN: 1234
Access granted.

2. Text Preprocessing: Skip Stopwords

#Problem:

#Parking fee calculation per hour

```
hourly_rate = 50
```

```
hours = int(input("Enter number of hours parked: "))
```

```
total = 0
```

```
while hours > 0:
```

```
    total += hourly_rate
```

```
    hours -= 1
```

```
print(f"Total parking fee: Rs {total}")
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
Microsoft Windows [Version 10.0.22631.5624]  
(c) Microsoft Corporation. All rights reserved.
```

```
C:\Users\Dr Imtiaz Korejo\Desktop\pythonworkspace>
```

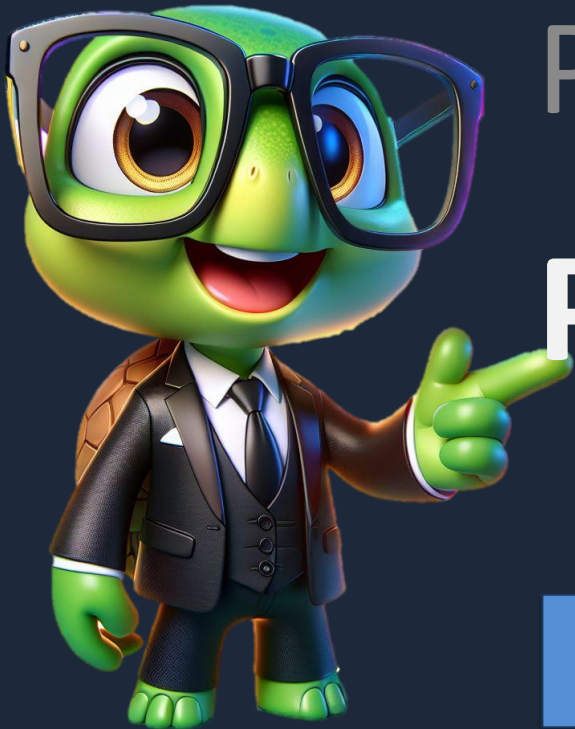
```
Enter number of hours parked: 5
```

```
Total parking fee: Rs 250
```

Section-3

Python Programming

PYTHON FOR LOOP



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Python For Loop

- A **for** loop is used for iterating over a sequence (that is either a list, a tuple, a dictionary, a set, or a string).
- This is less like the **for** keyword in other programming languages, and works more like an iterator method as found in other object-orientated programming languages.
- The **for** loop does not require an indexing variable to set beforehand.

```
File Edit Selection View Go Run ...  
Example_2.py Example_3 Example_4.py Example_5.py  
fruits = ["apple", "banana", "cherry"]  
  
for x in fruits:  
    print(x)  
  
Microsoft Windows [Version 10.0.22631.5624]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\Dr Imtiaz Korejo\Desktop\pythonworkspace>  
  
apple  
banana  
cherry
```





Looping through a String

- Even strings are iterable objects, they contain a sequence of characters:

Loop through the letters in the word "banana":

```
File Edit Selection View Go Run ... < >
Example_2.py Example_3 Example_4.py Example_5.py

for x in "banana":
    Print(x)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Microsoft Windows [Version 10.0.22631.5624]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Dr Imtiaz Korejo\Desktop\pythonworkspace>

b
a
n
a
n
a



The range() Function

- To loop through a set of code a specified number of times, we can use the `range()` function,
- The `range()` function returns a sequence of numbers, starting from 0 by default, and increments by 1 (by default), and ends at a specified number.

```
File Edit Selection View Go Run ... ← →  
Example_2.py Example_3 × Example_4.py Example_5.py  
  
for x in range(6):  
    print(x)  
  
Microsoft Windows [Version 10.0.22631.5624]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\Dr Imtiaz Korejo\Desktop\pythonworkspace>  
0  
1  
2  
3  
4  
5
```



Range() Function

- The `range()` function defaults to 0 as a starting value, however it is possible to specify the starting value by adding a parameter: `range(2, 6)`, which means values from 2 to 6 (but not including 6):

```
File Edit Selection View Go Run ... ← →
Example_2.py Example_3 × Example_4.py Example_5.py

for x in range(2, 6):
    print(x)

Microsoft Windows [Version 10.0.22631.5624]
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C:\Users\Dr Imtiaz Korejo\Desktop\pythonworkspace>

2
3
4
5
```



Range() Function Increment

- The `range()` function defaults to increment the sequence by 1, however it is possible to specify the increment value by adding a third parameter: `range(2, 30, 3)`:

```
File Edit Selection View Go Run ...  
Example_2.py Example_3 Example_4.py Example_5.py  
#Increment the sequence with 3 (default is 1):  
  
for x in range(2, 20, 3):  
    print(x)  
  
Microsoft Windows [Version 10.0.22631.5624]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\Dr Imtiaz Korejo\Desktop\pythonworkspace>  
  
2  
5  
8  
11  
14  
17  
20
```





The Break Statement

- With the **break** statement we can stop the loop before it has looped through all the items:
- Exit the loop when i is 5:

```
File Edit Selection View Go Run ... ← →  
Example_2.py Example_3 × Example_4.py Example_5.py  
for i in range(1, 10):  
    if i == 5:  
        print("Found 5, stopping loop.")  
        break  
    print(i)  
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS  
Microsoft Windows [Version 10.0.22631.5624]  
(c) Microsoft Corporation. All rights reserved.  
C:\Users\Dr Imtiaz Korejo\Desktop\pythonworkspace>  
1  
2  
3  
4  
Found 5, stopping loop
```



The Continue Statement

- With the **continue** statement we can stop the current iteration of the loop, and continue with the next:

Do not print Odd numbers:

```
File Edit Selection View Go Run ... ← →  
Example_2.py Example_3 × Example_4.py Example_5.py  
#Printing even numbers only  
for i in range(1, 11):  
    if i % 2 != 0: # Skip odd numbers  
        continue  
    print(f"Even number: {i}")  
  
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS  
Microsoft Windows [Version 10.0.22631.5624]  
(c) Microsoft Corporation. All rights reserved.  
C:\Users\Dr Imtiaz Korejo\Desktop\pythonworkspace>  
Even number: 2  
Even number: 4  
Even number: 6  
Even number: 8  
Even number: 10
```





Else in For Loop

- The else keyword in a for loop specifies a block of code to be executed when the loop is finished:

```
File Edit Selection View Go Run ... ← →  
Example_2.py Example_3 × Example_4.py Example_5.py  
#Print all numbers from 0 to 5, and print a message when the loop has ended:  
  
for x in range(6):  
    print(x)  
else:  
    print("Finally finished!")  
  
Microsoft Windows [Version 10.0.22631.5624]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\Dr Imtiaz Korejo\Desktop\pythonworkspace>  
0  
1  
2  
3  
4  
5  
Finally finished!
```



Nested Loops

- A nested loop is a loop inside a loop.
- The "inner loop" will be executed one time for each iteration of the "outer loop":

```
File Edit Selection View Go Run ... ← →  
Example_2.py Example_3 × Example_4.py Example_5.py  
  
for i in range(1, 4): # Outer loop → table number  
    for j in range(1, 6): # Inner loop → multiply by 1–5  
        print(f"{i} x {j} = {i*j}")  
    print("---")  
  
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS  
  
Microsoft Windows [Version 10.0.22631.5624]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\Dr Imtiaz Korejo\Desktop\pythonworkspace>  
  
1 x 1 = 1  
...  
1 x 5 = 5  
---  
2 x 1 = 1  
...  
2 x 5 = 10  
---  
3 x 1 = 3  
...  
3 x 5 = 15  
---
```





The pass Statement

- `for` loops cannot be empty, but if you for some reason have a `for` loop with no content, put in the `pass` statement to avoid getting an error.

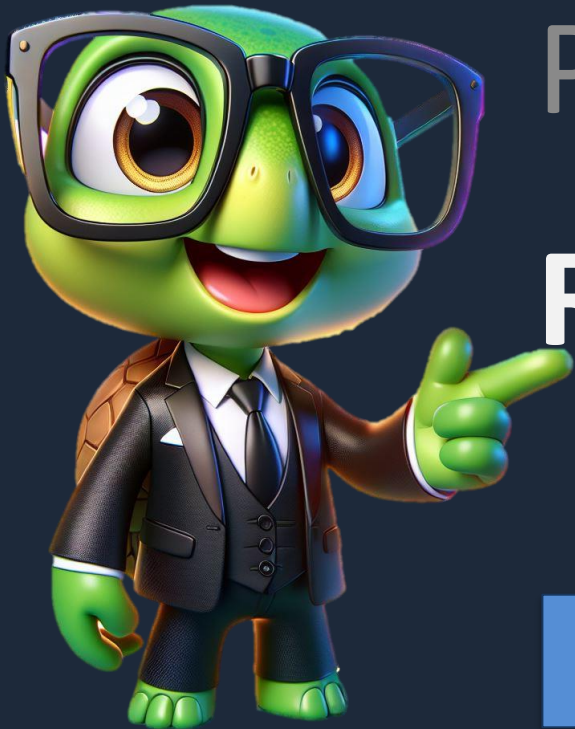
```
File Edit Selection View Go Run ...  
Example_2.py Example_3 Example_4.py Example_5.py  
  
for x in [0, 1, 2]:  
    pass  
  
# having an empty for loop like this, would raise an error without  
the pass statement  
  
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS  
Microsoft Windows [Version 10.0.22631.5624]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\Dr Imtiaz Korejo\Desktop\pythonworkspace>
```



Section-3

Python Programming

REAL WORLD EXAMPLES



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1. Weather Data Checker

Problem:

- Create an app that loops through a list of cities and checks their temperature using a weather API.
- If a city returns an error (e.g., invalid city), skip using continue.
- Stop the loop using break if temperature exceeds 40°C.



Install Module “requests” by typing following command on terminal

`pip install requests`

```
Microsoft Windows [Version 10.0.22631.5624]  
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```

```
C:\Users\Dr Imtiaz Korejo\Desktop\pythonworkspace>
```

```
pip install requests
```

```
Collecting requests
```

```
  Downloading requests-2.32.3-py3-none-any.whl.metadata (4.6 kB)
```

```
Collecting charset-normalizer<4,>=2 (from requests)
```

```
  Downloading charset_normalizer-3.4.1-cp312-cp312-macosx_10_13_universal2.whl.metadata (35 kB)
```

```
Collecting idna<4,>=2.5 (from requests)
```

```
  Downloading idna-3.10-py3-none-any.whl.metadata (10 kB)
```

```
Collecting urllib3<3,>=1.21.1 (from requests)
```

```
  Downloading urllib3-2.3.0-py3-none-any.whl.metadata (6.5 kB)
```

```
Collecting certifi>=2017.4.17 (from requests)
```

```
  Downloading certifi-2025.1.31-py3-none-any.whl.metadata (2.5 kB)
```

```
Downloading requests-2.32.3-py3-none-any.whl (64 kB)
```

```
Downloading certifi-2025.1.31-py3-none-any.whl (166 kB)
```

```
Downloading charset_normalizer-3.4.1-cp312-cp312-macosx_10_13_universal2.whl (196 kB)
```

```
Downloading idna-3.10-py3-none-any.whl (70 kB)
```

```
Downloading urllib3-2.3.0-py3-none-any.whl (128 kB)
```

```
Installing collected packages: urllib3, idna, charset-normalizer, certifi, requests
```

```
Successfully installed certifi-2025.1.31 charset-normalizer-3.4.1 idna-3.10 requests-2.32.3 urllib3-2.3.0
```




- 1. Visit <https://openweathermap.org/>
- Create Account.

The screenshot shows the OpenWeather website interface. At the top, there's a navigation bar with links like 'Guide', 'API', 'Dashboard', 'Marketplace', 'Pricing', 'Maps', 'Our Initiatives', 'Partners', 'Blog', 'For Business', 'Sign in', and 'Support centre'. Below the navigation bar is a large banner with the OpenWeather logo and the text 'Weather forecasts, nowcasts and history in a fast and elegant way'. A search bar is present with the placeholder text 'Search city' and a 'Search' button. Below the search bar, the weather data for 'Kotri, PK' is displayed for 'Apr 4, 06:10pm'. The current temperature is 39°C, with a 'Feels like 36°C. Clear sky. Light breeze'. Additional details include wind speed (2.8m/s NNW), pressure (1008hPa), humidity (5%), UV index (0), dew point (-6°C), and visibility (10.0km). A map of the region is shown, highlighting 'HYDERABAD' and 'SINDH'. A precipitation forecast for the next hour is displayed, stating 'No precipitation within an hour'. A chat button for 'Ulla Weather Assistant' is visible in the bottom right corner.

OpenWeather

Weather forecasts, nowcasts and history in a fast and elegant way

Search city Search

Apr 4, 06:10pm
Kotri, PK

● **39°C**

Feels like 36°C. Clear sky. Light breeze

▼ 2.8m/s NNW ☉ 1008hPa
Humidity: 5% UV: 0 Dew point: -6°C
Visibility: 10.0km

HYDERABAD SINDH

No precipitation within an hour

now 15min 30min 45min 60min
06:10pm 06:25pm 06:40pm 06:55pm 07:10pm

Ulla Weather Assistant



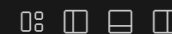


- Under Your Profile, click “My API Keys” and Copy the key or create new one.

The screenshot shows the OpenWeather API dashboard. The top navigation bar includes links like 'Weather in your city', 'Guide', 'API', 'Dashboard', 'Marketplace', 'Pricing', 'Maps', 'Our Initiatives', 'Partners', 'Blog', 'For Business', and a user profile 'zeesh...'. Below this, a secondary navigation bar has links for 'New Products', 'Services', 'API keys' (which is underlined), 'Billing plans', 'Payments', 'Block logs', 'My orders', 'My profile', and 'Ask a question'. A dropdown menu is open from the user profile, showing options: 'My services', 'My API keys' (highlighted with a red box), 'My payments', 'My profile', and 'Logout'. Below the navigation, a light blue box contains the text: 'You can generate as many API keys as needed for your subscription. We accumulate the total load from all of them.' Below this is a table with columns: 'Key', 'Name', 'Status', and 'Actions'. The table has two rows. The first row has a key '328d52778f8db45bb309b4655cd0119b', name 'Default', and status 'Active'. The second row has a key '8b86db894aa200ea3847f7e498731a98', name 'PythonClass', and status 'Active'. The second key is highlighted with a red box. To the right of the table is a 'Create key' section with a text input 'API key name' and a 'Generate' button.

Key	Name	Status	Actions
328d52778f8db45bb309b4655cd0119b	Default	Active	  
8b86db894aa200ea3847f7e498731a98	PythonClass	Active	  





Python-ClassExample.py

PythonExample.py ×



```
import requests
```



```
cities = ["Karachi", "Hyderabad", "InvalidCity", "Kotri", "Lahore"]
```



```
API_KEY = "8b86db894aa200ea3847f7e498731a98" # PASTE your OpenWeatherMap API Key here
```



```
for city in cities:
```



```
    try:
```



```
        response = requests.get( f"https://api.openweathermap.org/data/2.5/weather?q={city}&appid={API_KEY}&units=metric")
        data = response.json()
```

```
        if data.get("cod") != 200:
            print(f"Skipping {city} - Not Found")
            continue
```

```
        temp = data["main"]["temp"]
        print(f"{city}: {temp}°C")
```

```
        if temp > 40:
            print(f"Warning: High temperature in {city}. Stopping.")
            break
```

```
except Exception as e:
    print(f"Error with city {city}: {e}")
    continue
```





Python-ClassExample.py

PythonExample.py ×



```
import requests
```



```
cities = ["Karachi", "Hyderabad", "InvalidCity", "Kotri", "Lahore"]
```



```
API_KEY = "8b86db894aa200ea3847f7e498731a98" # PASTE your OpenWeatherMap API Key here
```



```
for city in cities:
```



```
    try:
```

```
        response = requests.get(f"https://api.openweathermap.org/data/2.5/weather?q={city}&appid={API_KEY}&units=metric")
        data = response.json()
```

```
        if data.get("cod") != 200:
            print(f"Skipping {city} - Not Found")
            continue
```

```
        temp = data["main"]["temp"]
        print(f"{city}: {temp}°C")
```

```
        if temp > 40:
            print(f"Warning: High temperature in {city}. Stopping.")
            break
```

```
    except Exception as e:
        print(f"Error with city {city}: {e}")
        continue
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Microsoft Windows [Version 10.0.22631.5624]
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```

```
C:\Users\Dr Imtiaz Korejo\Desktop\pythonworkspace>
```

```
Karachi: 30.08°C
Hyderabad: 32.17°C
Skipping InvalidCity - Not Found
Kotri: 39.28°C
Lahore: 31.8°C
```

Programming for AI

NEXT CLASS...

MORE CODING EXAMPLES



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Programming for AI

END of SESSION

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Week- , Lecture-

